

PPF and the opportunity cost

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ECON 101- Fall 2020

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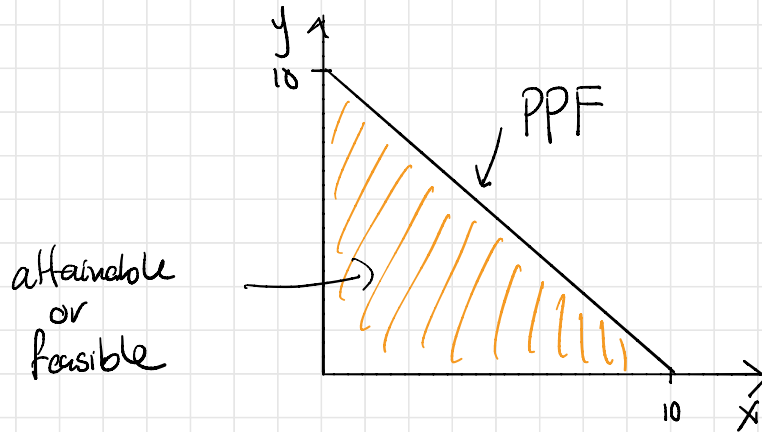


In general, the production possibilities set  $X$

$X = \{ \text{set of all output combinations that are feasible, given current technology} \}$

Examples:

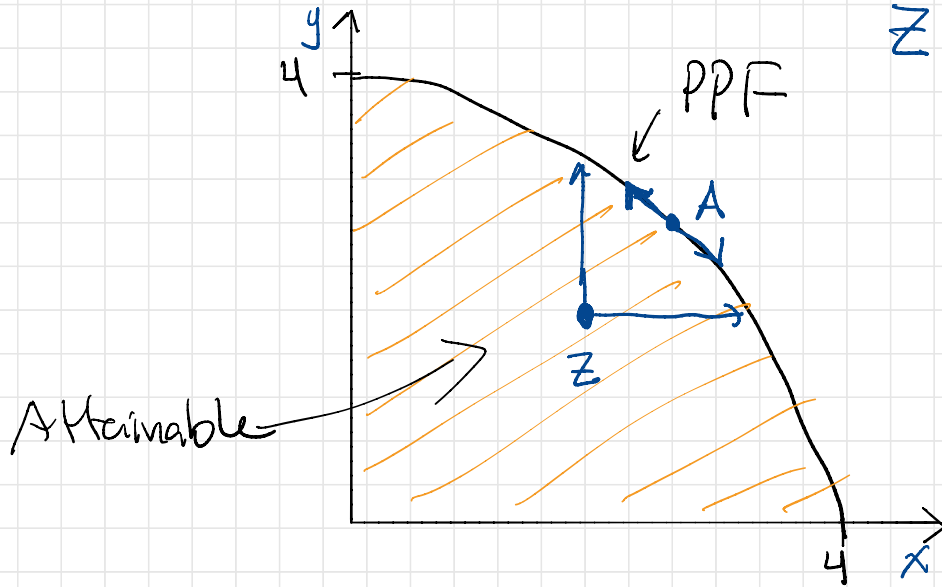
Linear Production Possibilities Set:  $X + y \leq 10$



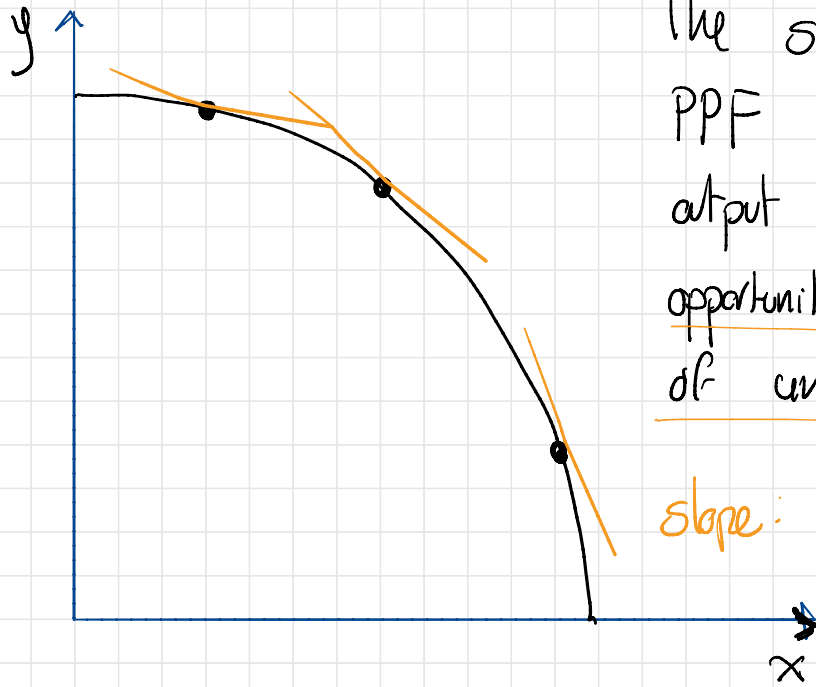
Another example:

Concave Production Possibilities Set

$$x^2 + y^2 \leq 16$$



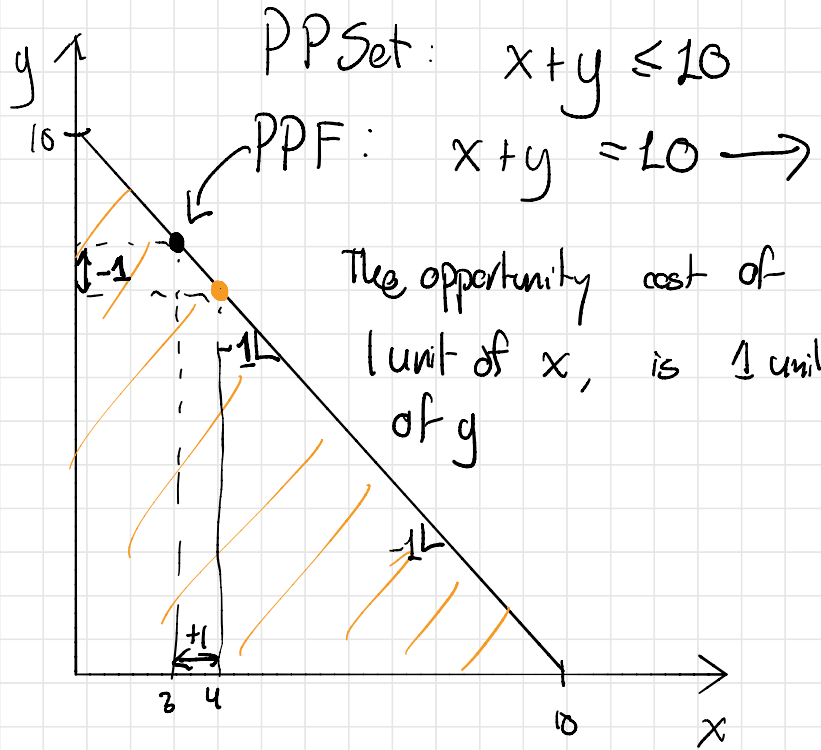
Z is not efficient!



The slope of the the PPF at a particular output combination is the opportunity cost of  $x$  in terms of units of  $y$  forgone

slope:  $OC$  Limit  $x$  in terms of units of  $y$  forgone

Math Reminder: the slope of a curve at a point  $(x, y)$  is the same as the slope of the tangent line at that point.



$y = 10 - x$  ← slope = -1

Reminder:

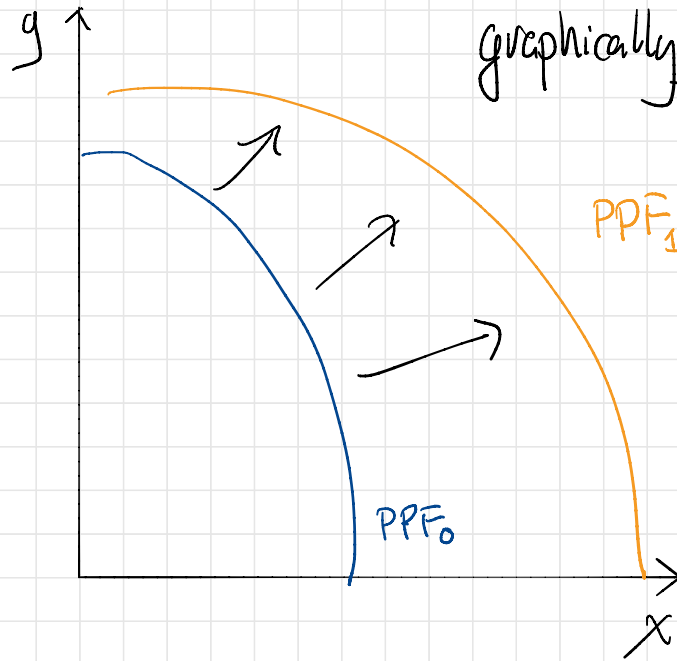
The general eq. of a line in the plane:

$$y = b + m x$$

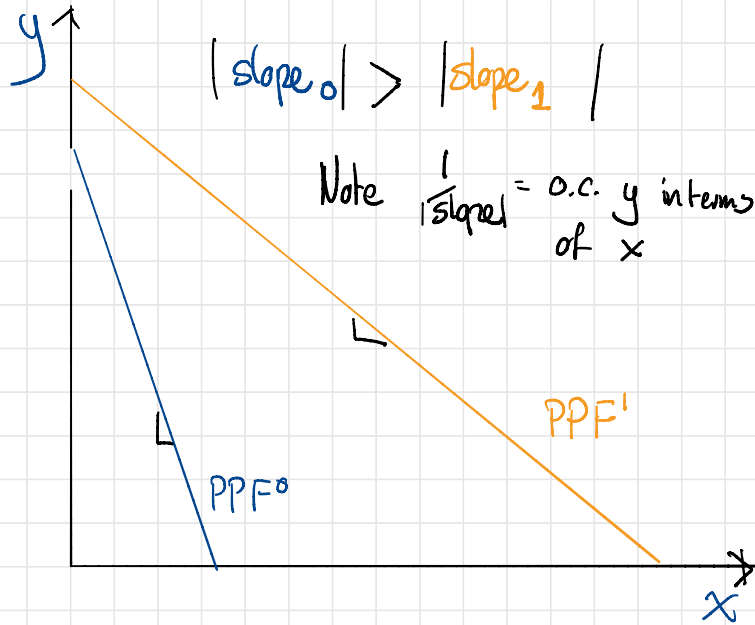
$\uparrow$                        $\uparrow$   
 vertical                  slope  
 intercept

For linear PPFs, the slope is constant  
 $\rightarrow$  the opportunity cost is constant

# Economic Growth,



graphically, an expansion of the PPF outwards represents growth



1 has a C.A. in  $x$   
 0 has a C.A. in  $y$ ,

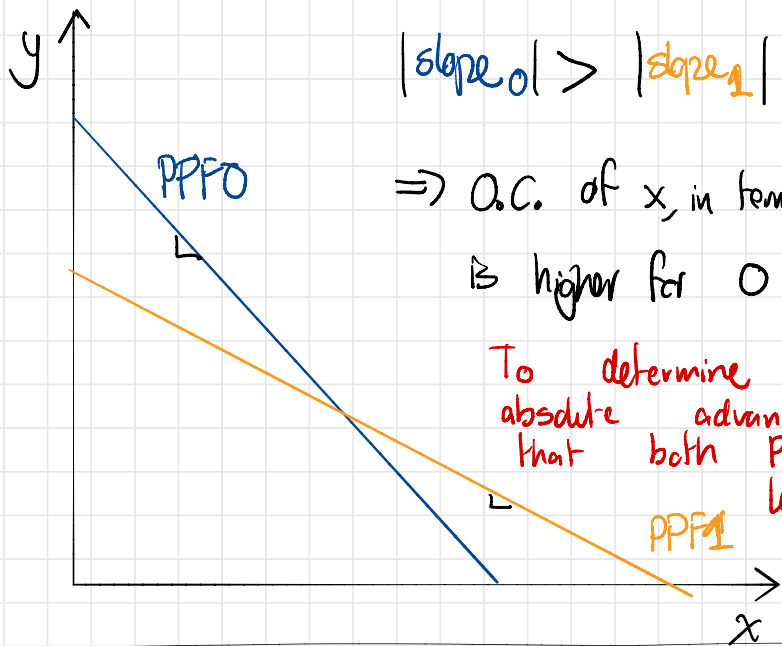
1, using the same resources,  
 is much more productive than  
 0  $\Rightarrow$  1 has an absolute  
 advantage in the production  
 of  $x$  and  $y$

Comparative and  
 Absolute advantage

For  $PPF_0$  the opportunity cost of  $x$  is  
 higher than for  $PPF_1$ : 1 has a comparative  
 advantage in the production of good  $x$ , 0 has C.A. in production of  $y$

Comparative advantage but we don't have enough information to determine if there is absolute advantage.

↳ We don't have information about inputs of production



$$|\text{slope}_0| > |\text{slope}_1|$$

$\Rightarrow$  O.C. of  $x$ , in terms of  $y$   
is higher for 0

1 has C.A. in  $x$   
0 has C.A. in  $y$

To determine if somebody has an absolute advantage we need to know that both PPF's had access to the same level of inputs!

MORAL: • Comparative advantage always exist, and can be glanced only from the differences in slopes

• For comparative advantage we need also information on whether both PPF's had access to the SAME level of inputs,

**Example:**  $x = \text{smoothies}$   
 $y = \text{salads}$

60 min, Joe can produce  
10 smoothies, OR 2 salads

$$60 = 10x + 2y; y = 30 - 5x$$

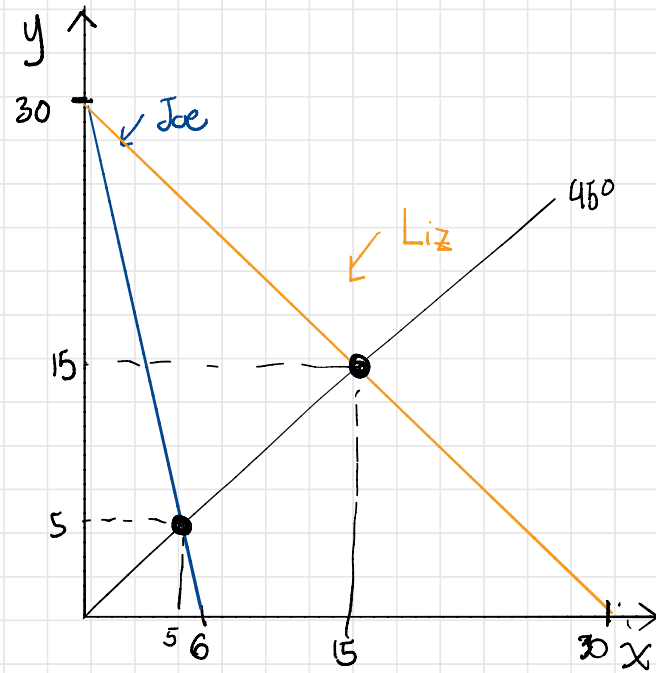
Cost of opportunity of smoothies for Joe: 5 salads

60 minutes, Liz can produce  
30 smoothies / 30 salads

She takes 2 minutes to produce either of them

$$60 = 2x + 2y; y = 30 - x$$

Cost of opportunity of smoothies for Liz: 1 salad.



Liz has a comparative advantage  
in smoothies; Joe has a  
C.A in salad production;  
Liz has an absolute advantage in  
smoothie production